comparison. (3) *Pause as thought*: We use special <pause> tokens to replace the continuous thoughts, and apply the same multi-stage training curriculum as COCONUT.

5.3 Results and Discussion

We show the overall results in Table 1. Using continuous thoughts effectively enhances LLM reasoning over the No-CoT baseline. For example, by using 6 continuous thoughts, COCONUT achieves 34.1% accuracy on GSM8k, which significantly outperforms *No-CoT* (16.5%). We highlight several key findings below.

"Chaining" continuous thoughts enhances reasoning. Language CoT proves to increase the effective depth of LLMs and enhance their expressiveness (Feng et al., 2023). Thus, generating more tokens serves as a way to inference-time scaling for reasoning (Guo et al., 2025; Snell et al., 2024). This desirable property holds naturally for COCONUT too. On GSM8k, COCONUT outperformed other architectures trained with similar strategies, including COCONUT (*pause as thought*) and COCONUT (*w/o thought*). Particularly, it surpasses the latest baseline *iCoT* (Deng et al., 2024), which requires a more carefully designed training schedule.

Additionally, we experimented with adjusting the hyperparameter c, which controls the number of latent thoughts corresponding to one language reasoning step (Figure 8, II). As we increased c from 0 to 1 to 2, the model's performance steadily improved.[2] This further validates the potential of continuous thoughts to scale up to harder problems. In two other synthetic tasks, we found that the variants of COCONUT (*w/o thoughts* or *pause as thought*), and the *iCoT* baseline also achieve impressive accuracy. This indicates that the model's computational capacity may not be the bottleneck in these tasks. In contrast, GSM8k involves more complex contextual understanding and modeling, placing higher demands on computational capability.

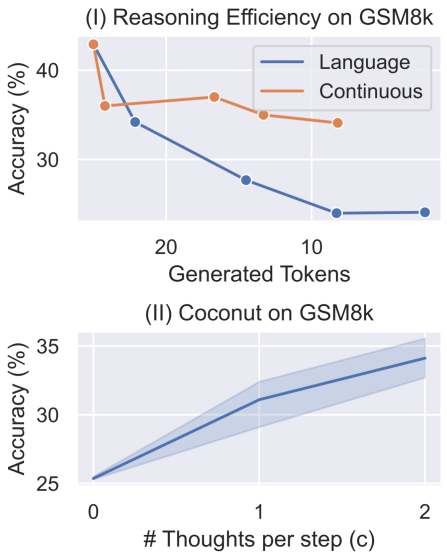

Figure 8 Efficiency comparison of reasoning space and COCONUT with different c.

Continuous thoughts are efficient representations of reasoning. Compared to traditional CoT, COCONUT generates fewer tokens while achieving higher accuracy on ProntoQA and ProsQA (Table 1). Although COCONUT does not surpass *CoT* on GSM8k, it offers a superior trade-off between reasoning efficiency and accuracy (Figure 8, I). To illustrate this, we train a series of CoT models that progressively "internalize" (Deng et al., 2024) the initial $m = \{0, 1, 2, 3, \text{ALL}\}$ reasoning steps, and plot their accuracy versus the number of generated tokens (labeled as "language" in the figure). These models quickly lose accuracy as more reasoning steps are skipped. In contrast, by applying COCONUT training strategy—replacing each language reasoning step with two continuous thoughts—the accuracy drop is substantially mitigated, maintaining higher performance even when fewer tokens are generated. Another interesting observation is that, when we decode the first continuous thought, it often corresponds to possible intermediate variables in the calculation (Figure 9). This also suggests that the continuous thoughts are more efficient representations of reasoning.

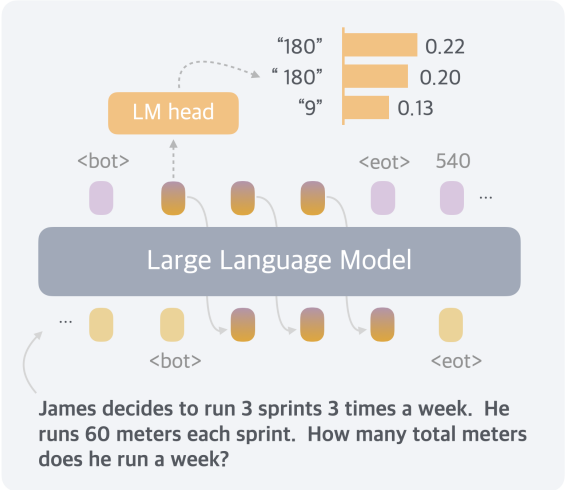

Figure 9 Decoding a continuous thought into language tokens in a math word problem. The decoded tokens correspond to intermediate variables that help solve the problem.

[2]We discuss the case of larger c in Appendix C.1.

The LLM still needs guidance to learn latent reasoning. In the ideal case, the model should learn the most effective continuous thoughts automatically through gradient descent on questions and answers (i.e., COCONUT *w/o curriculum*). However, from the experimental results, we found the models trained this way do not perform any better than no-CoT.

On the contrary, with the multi-stage curriculum, COCONUT is able to achieve top performance across various tasks. The multi-stage training also integrates well with pause tokens (COCONUT- *pause as thought*). Despite using the same architecture and similar multi-stage training objectives, we observed a small gap between the performance of *iCoT* and COCONUT (*w/o thoughts*). The finer-grained removal schedule (token by token) and a few other tricks in *iCoT* may ease the training process. We leave combining *iCoT* and COCONUT as future work. While the multi-stage training used for COCONUT has proven effective, further research is definitely needed to develop better and more general strategies for learning reasoning in latent space, especially without the supervision from language reasoning chains.

6 Conclusion

In this paper, we introduce COCONUT, a new paradigm for reasoning in continuous latent space. Experiments demonstrate that COCONUT effectively enhances LLM performance across a variety of reasoning tasks. Reasoning in latent space gives rise to advanced emergent behaviors, where continuous thoughts can represent multiple alternative next steps. This enables the model to perform BFS over possible reasoning paths, rather than prematurely committing to a single deterministic trajectory as in language space CoT reasoning. Further research is needed to refine and scale latent reasoning to pretraining, which could improve generalization across a broader range of reasoning challenges. We hope our findings will spark continued exploration into latent reasoning, ultimately advancing the development of more capable machine reasoning systems.

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